Dr. Mukesh Singh

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Personal website
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Research Interests: My passion lies in understanding the fundamental properties of nanomaterials, including their electronic, optical, an-harmonic, and elastic characteristics, and predicting novel materials using ab initio methods and machine learning approaches. These attributes of material extend my interest in practical applications, such as thermoelectric properties, thermal conductivity, batteries (metal-ion and metal-sulfur), catalysis reactions like Hydrogen Evolution Reactions and Oxygen Evolution Reactions etc, and hydrogen storage.

Research Experience

Jan, 2025 – present	**Postdoctoral fellow,** Department of Physics, Central Michinga University, USA. Novel electrode materials using first principles methods, metal-sulphur ion battery.
April, 2024 – December, 2024	**Postdoctoral fellow,** Department of Physics, IIT Bombay. Quantum dot-modeling and properties (electronic and optical characterization); 2D materials-thermal conductivity and thermoelectric properties

Education

2019 – 2024	**Ph.D. (Computational Condensed Matter Physics)** defended on 29 April 2024. Indian Institute of Technology, Bombay, Mumbai, India. **Thesis title:** Structural and Electronic Properties of 2D Materials and their Applications in Hydrogen Production and Hydrogen Storage: First Principles Study.
2015 – 2017	**M.Sc. (Physics)** Kirori Mal College, Delhi University, India. **Thesis title:** Stringy Correction to Reisssner-Nordstrom Black Hole: Multi Universe Scenario.
2012-2015	**B.Sc. (Physics, Chemistry, Mathematics)** Deep Dayal Upadhyay Gorakhpur University, Gorakhpur, India.

Research Skills

Projects	Vestapy, RxnPy, VaspWorks, Pre_and_post_processing_dft (see github.com/mukesh4iitb)
Programming Language	Python, Fortran, Bash(awk,sed, etc.), C, C++ (beginner), Foundation of machine learning.
Material modeling Python libraries	Atomic Simulation Environment(ASE), Pymatgen
DFT codes	VASP, Gaussian, Quantum Espresso, aflow
Other utility codes	Phonopy, Phono3py, ShengBTE/almaBTE, Boltztrap2

Research Skills (continued)

Misc. LaTeX, lyx and libreoffice-suite, VESTA, XCrysden, GIMP, Inkscape, Origin-lab.

Research Publications

Journal Articles

1. **M. Singh**, S. P. Kaur, and B. Chakraborty, "Modeling and tuning the electronic, mechanical and optical properties of a recently synthesized 2d polyaramid: A first principles study," *Phys. Chem. Chem. Phys*, vol. 26, p. 21 874, Jul. 2024. DOI: 10.1039/D4CP02027H
2. **M. Singh**, A. Shukla, and B. Chakraborty, "Improving hydrogen evolution catalytic activity of 2D carbon allotrope biphenylene with B, N, P doping: Density functional theory investigations," *International Journal of Hydrogen Energy*, pp. 569–579, 2024. DOI: https://doi.org/10.1016/j.ijhydene.2023.08.359
3. P. Mane, S. P. Kaur, **M. Singh**, A. Kundu, and B. Chakraborty, "Superior hydrogen storage capacity of vanadium decorated biphenylene (Bi+V): A DFT study," *International Journal of Hydrogen Energy*, vol. 48, pp. 28 076–28 090, 2023. DOI: https://doi.org/10.1016/j.ijhydene.2023.04.033
4. **M. Singh** and B. Chakraborty, "2D BN -biphenylene: Structure stability and properties tenability from a DFT perspective," *Phys. Chem. Chem. Phys.*, vol. 25, pp. 16 018–16 029, 2023. DOI: 10.1039/D3CP00776F
5. **M. Singh**, A. Shukla, and B. Chakraborty, "High capacity hydrogen storage on zirconium decorated γ-graphyne: A systematic first-principles study," *International Journal of Hydrogen Energy*, vol. 48, pp. 37 834–37 846, 2023. DOI: https://doi.org/10.1016/j.ijhydene.2022.07.062
6. **M. Singh**, A. Shukla, and B. Chakraborty, "Highly efficient hydrogen storage of a Sc decorated biphenylene monolayer near ambient temperature: Ab initio simulations," *Sustainable Energy Fuels*, vol. 7, pp. 996–1010, 2023. DOI: 10.1039/D2SE01351G
7. **M. Singh**, A. Shukla, and B. Chakraborty, "An ab-initio study of the Y decorated 2D holey graphyne for hydrogen storage application," *Nanotechnology*, vol. 33, p. 405 406, Jul. 2022. DOI: 10.1088/1361-6528/ac7cf6

Sumitted Manuscripts

1. J. Pandey, A. Yazdani, **M. Singh**, .., and B. D. Fahlman, *Regulating polysulfide conversion on Fe–N_x single-atom catalysts supported on graphitic carbon nitride revealed by semi-in situ XPS of Li-S batteries.*
2. **M. Singh** and V. Barone, *Unified thermodynamic and kinetic criteria for screening 2d anchoring materials in Li–S, Na–S, and K–S batteries.*

Manuscripts under Preparation

1. **M. Singh**, V. Sharma, and A. Shukla, *Electrical and optical properties of novel tetra-penta-deca-hexa (TPDH) graphene quantum dot.*
2. **M. Singh**, S. S. Shastri, and A. Shukla, *Lattice dynamics and thermal conductivity of 2D monolayer MH_2: A first principles study.*

Workshops and conferences

1. **M. Singh** and V. Barone, *Computational investigation of potassium–sulfur electrode performance on nitrogen-rich carbon supports*, APS Global Physics Summit, Denver, Colorado (Boulder), USA, **Contributed talk**, Mar. 2026.
2. **M. Singh**, M. Jhakar, and V. Barone, *Single atom catalysts (co/fe/ni) adsorbed on graphitic carbon nitride for potassium–sulfur batteries*, SUNCAT Summer Institute, SLAC Stanford University, Stanford, California, USA, **Poster presentation**, Aug. 2025.
3. **M. Singh**, *2D BN-Biphenylene: Structure stability and properties tunability with DFT perspective*, Vienna Ab-initio Simulation Package (VASP) and Applications, Online, **Contributing talk**, Feb. 2024.
4. **M. Singh**, S. S. Shastri, and A. Shukla, *Lattice dynamics and thermal conductivity of 2d monolayer MH2 and their application in hydrogen vehicles: A first-principles study*, International Conference on 60 Years of DFT: Advancements in Theory & Computation, held at IIT Mandi, India, **Presented Poster**, Jul. 2024.
5. **M. Singh**, *An ab - initio study of y decorated holey graphyne and zr decorated graphyne for hydrogen storage application*, Conference on Advances of Renewable Energy (CARE-2023), held at HRI, Prayagraj, India, **Presented Poster**, Feb. 2023.
6. **M. Singh**, *Improving hydrogen evolution catalytic activity of 2D carbon allotrope Biphenylene with B, N, P doping: Density functional theory investigations*, 2022 World Hydrogen Energy Conference, Online, **Presented Paper**, May 2023.
7. **M. Singh**, *Structural, electronic and optical properties of graphene like dhq-qd: Confinement effect*, 2nd International meeting on Energy Storage devices (IMESD)-2023 & Industrial academic conclave, held at IIT Roorkee, Uttarakhand, India, **Presented Poster**, Dec. 2023.
8. **M. Singh** and A. Shukla, *Highly efficient hydrogen storage of Sc decorated biphenylene monolayer near ambient-temperature: Ab-initio simulations*, SymPhy 2023, Online, **Presented H2 storage works**, Feb. 2023.
9. **M. Singh**, *High capacity hydrogen storage in zirconium functionalized γ-graphyne: A systematic first-principles study*, International Conference On Renewable Energy, Online, **Presented Poster**, Feb. 2022.
10. **M. Singh**, V. Sharma, and A. Shukla, *Electrical and optical properties of novel tetra-penta-deca-hexa (TPDH) graphene quantum dot*, Workshop on Frontiers in Excited State Electronic Structure Methods: from Spectroscopy to Photochemistry, held at ICTP, Italy, **Presented Poster**, Jul. 2022.

Additional Professional Experience

Awards and Achievements

- 2021 **CSIR-SRF**, Council of Scientific and Industrial Research-Senior Research Fellow.
- 2018 **CSIR-JRF**, Council of Scientific and Industrial Research-Junior Research Fellow.
- 2017 **CSIR-JRF**, Council of Scientific and Industrial Research-Junior Research Fellow.
- 2015 **SELECTED in top 1%**, National Graduate Physics Examination-2015.
- 2014 **SELECTED in top 10%**, National Graduate Physics Examination-2014.
- 2013 **SELECTED in top 10%**, National Graduate Physics Examination-2013.
- 2012 **SELECTED in top 1%**, Innovation in Science Pursuit for Inspired Research (INSPIRE) Fellowship-2012.

Teaching assistant

- Advance simulation course with python (PH-413).
- Electricity and Magnetism (PH-108)
- Introductory and Advanced Simulation course with Fortran (PH-434, PH-566)
- Thermal Conductivity By Lee's Disc (PH-101)

Positions of representative (POR)

- 2020-2021 Academic Unit Representative for Academic Affairs (AURAA).
- 2021-2022 Academic Unit Representative for Academic Affairs (AURAA)
- 2020-2021 Research Scholars' Association of Physics (RSAP) General Secretary.
- 2022-2023 System admin H14 IIT Bombay.

References

Prof. Alok Shukla
(Ph.D. Supervisor)
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Prof. Brahmananda Chakraborty
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